



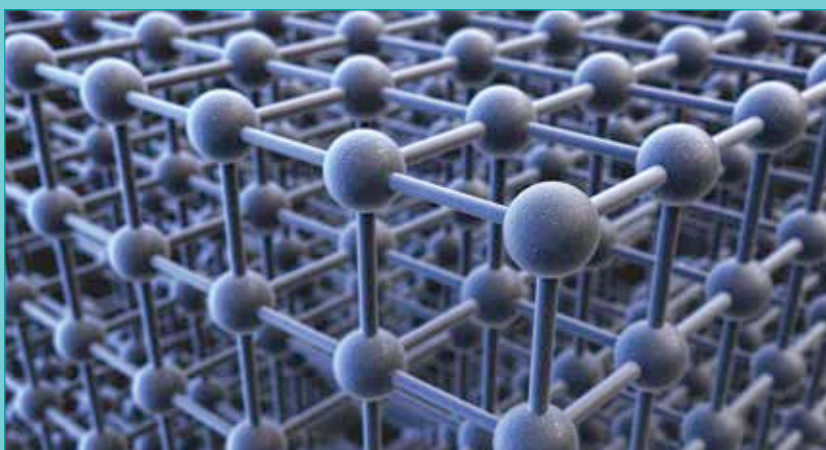
Global methane map

Canadian startup, GHGSat, has promised to release a high resolution world methane map. <https://digit.in/feb20-33>



NASA launches snow study

NASA began a snow study for the first time in 30 years to understand how snow bands form and evolve. <https://digit.in/feb20-34>



↑ The geometric appearance of crystals is because of the periodic arrangement of the individual atoms, ions or molecules. This structure is called the crystal lattice and the chemical properties of the constituent material decides how the lattice will be formed. The distance between two individual atoms, ions or molecules, the angle at which they link up are called lattice constants and based on all the three-dimensional possibilities, there are 230 possible groups or combinations. These space groups fall under different classifications based on the parameters used. These could be crystal system, crystal family or lattice system. We're looking at just crystal systems, which are seven in total. Each of which has a unique appearance.



↑ Orthorhombic - Tanzanite



↑ Monoclinic - Azurite



↑ Isometric/Cubic - Fluorite



↑ Tetragonal - Wulfenite



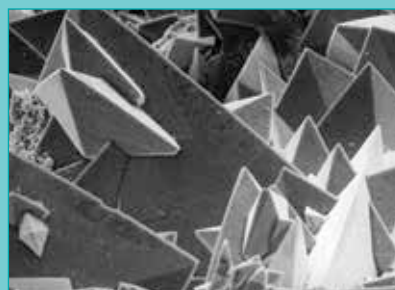
↑ Triclinic - Turquoise



↑ Hexagonal - Emerald



↑ Trigonal - Rhodochrosite



↑ Tetragonal crystal structure observed on the surface of a kidney stone



Credit: MIT

← The chemical and physical properties of different crystalline forms enable for different use cases and even allows for producing the same object with varying efficiency. You can produce solar cells using anything from purely crystalline to amorphous silicon. However, the cost also goes up significantly which is why you get cheap solar cells made from amorphous silicon whereas the high-performance solar cells cost a lot. Single-junction lab efficiency of monocrystalline silicon is 26.7 per cent and that's the highest conversion efficiency from among all the commercial photovoltaic technologies. Polycrystalline silicon comes close at 22.3 per cent and amorphous silicon solar cells have an abysmal 10.2 per cent conversion efficiency. Here's a polycrystalline silicon wafer coated with Silicon Nitride that's easy to produce and has a high conversion efficiency allowing for mass producing good solar cells.